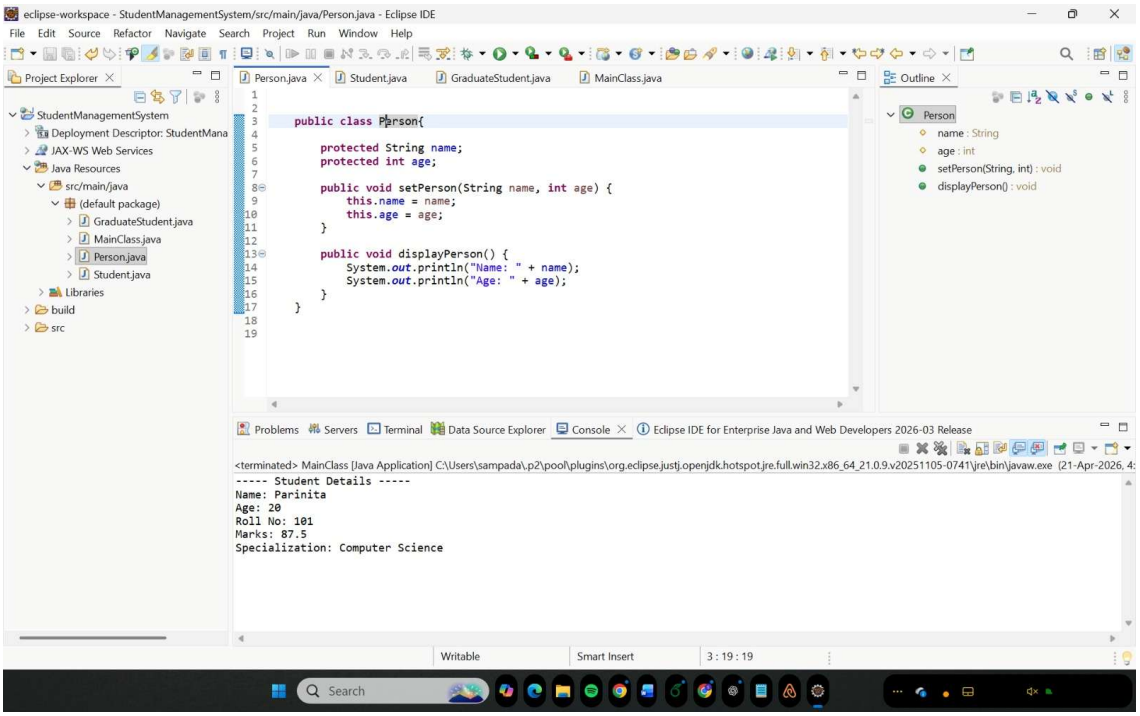
Flexi Credit Course - Java Case Study

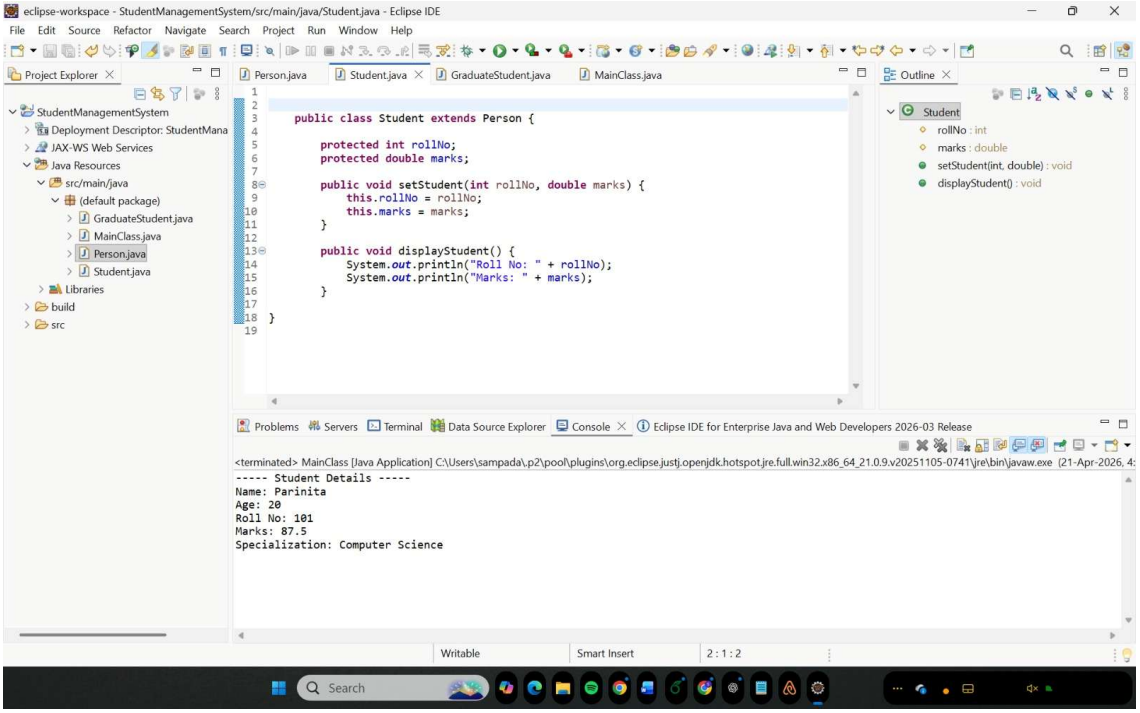
1. Student Management System

Problem Definition

Design a Java program to manage student information using inheritance. Create a base class Person that stores common details like name and age. Derive a class Student that includes additional attributes such as roll number and marks. Further extend it into another class GraduateStudent with specialization details. The program should display complete student information using inherited properties.




```
public class Person {
    protected String name;
    protected int age;
    public void setPerson(String name, int age) {
        this.name = name;
        this.age = age;
    }
    public void displayPerson() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }
}
```

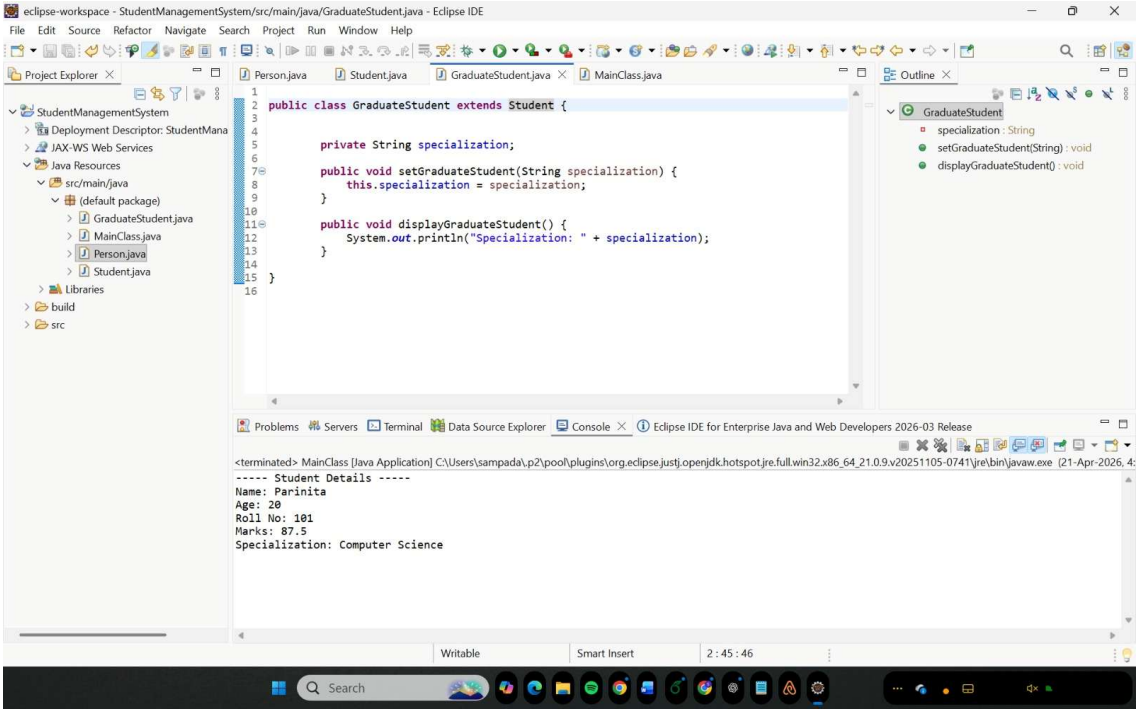



```
public class Student extends Person {

    protected int rollNo;
    protected double marks;

    public void setStudent(int rollNo, double marks) {
        this.rollNo = rollNo;
        this.marks = marks;
    }

    public void displayStudent() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Marks: " + marks);
    }
}
```

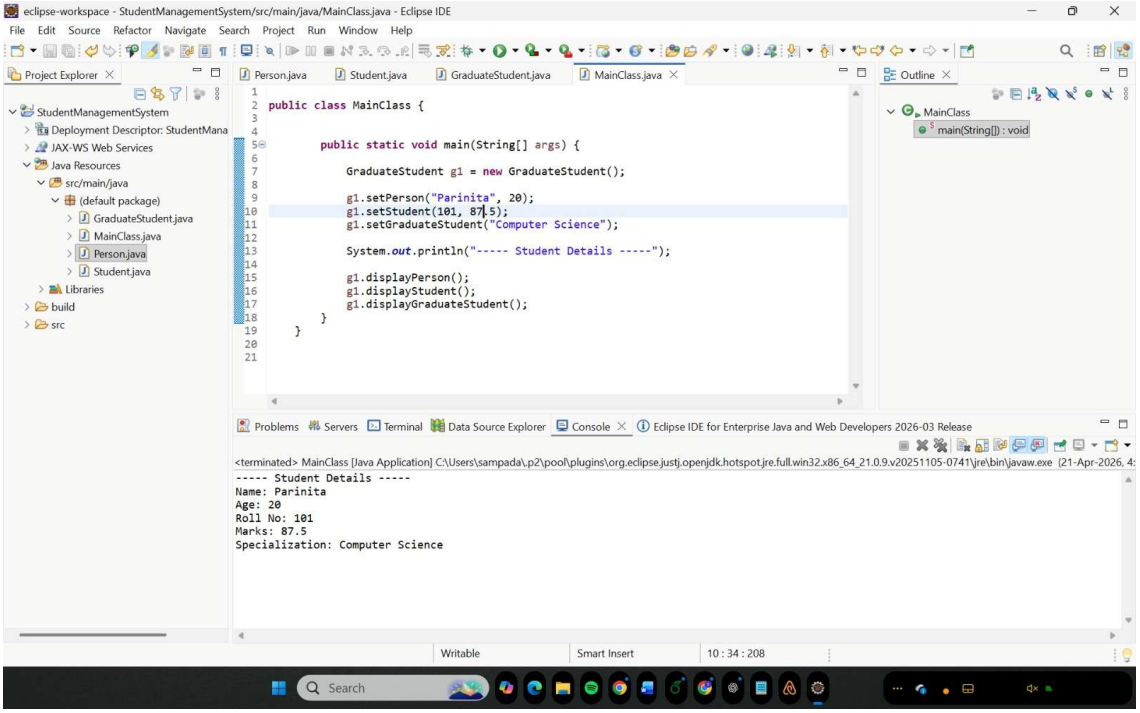



```
public class GraduateStudent extends Student {

    private String specialization;

    public void setGraduateStudent(String specialization) {
        this.specialization = specialization;
    }

    public void displayGraduateStudent() {
        System.out.println("Specialization: " + specialization);
    }
}
```




```
public class MainClass {
    public static void main(String[] args) {
        GraduateStudent g1 = new GraduateStudent();
        g1.setPerson("Parinita", 20);
        g1.setStudent(101, 85.5);
        g1.setGraduateStudent("Computer Science");
        System.out.println("----- Student Details -----");
        g1.displayPerson();
        g1.displayStudent();
        g1.displayGraduateStudent();
    }
}
```

Conclusion:

The Student Management System was successfully implemented using the concept of multilevel inheritance in Java. A base class Person was created to store common attributes like name and age. The Student class extended Person by adding roll number and marks, while GraduateStudent further extended Student to include specialization details.